

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

# Chapter 9: Matrices

Topic-wise short notes with solved examples

A matrix is a rectangular arrangement of numbers. This chapter covers the order and types of matrices, adding and subtracting them, multiplying by a number, and multiplying two matrices.

## 1 What a matrix is

- A **matrix** is a rectangular arrangement of numbers in rows and columns, enclosed in brackets. Each number is an **element**.
- A matrix with  $m$  rows and  $n$  columns has **order**  $m \times n$  (read " $m$  by  $n$ ") and  $m \times n$  elements. Rows are counted first.

$$A = \begin{bmatrix} 2 & 5 & -1 \\ 0 & 3 & 4 \end{bmatrix} \text{ has 2 rows and 3 columns, so its order is } 2 \times 3.$$

- Two matrices are **equal** if they have the same order and every pair of corresponding elements is equal.

| Type                       | Description                                    | Example                                        |
|----------------------------|------------------------------------------------|------------------------------------------------|
| Row matrix                 | Only one row                                   | $[3 \ 1 \ 4]$                                  |
| Column matrix              | Only one column                                | $\begin{bmatrix} 2 \\ 7 \end{bmatrix}$         |
| Square matrix              | Rows = columns                                 | $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ |
| Zero (null) matrix         | Every element is 0                             | $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ |
| Identity (unit) matrix $I$ | Square; 1 on the leading diagonal, 0 elsewhere | $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ |

- The **transpose**  $A^t$  is obtained by interchanging the rows and columns of  $A$ .

**Example 1.1** Find  $x$  and  $y$  if  $\begin{bmatrix} x+y & 5 \\ 2 & x-y \end{bmatrix} = \begin{bmatrix} 7 & 5 \\ 2 & 3 \end{bmatrix}$ .

**SOLUTION**

Equal matrices have equal corresponding elements:  $x + y = 7$  and  $x - y = 3$

Adding:  $2x = 10$

$$x = 5, y = 2$$

## 2 Addition, subtraction and multiplication by a number

- Two matrices can be added or subtracted only if they have the **same order**. Add or subtract the corresponding elements.
- To multiply a matrix by a number (a **scalar**), multiply every element by that number.
- $A + B = B + A$ , and  $A + O = A$  where  $O$  is the zero matrix of the same order.

**Example 2.1** If  $A = \begin{bmatrix} 2 & -1 \\ 3 & 4 \end{bmatrix}$  and  $B = \begin{bmatrix} 1 & 5 \\ -2 & 0 \end{bmatrix}$ , find  $A + B$  and  $2A - 3B$ .

SOLUTION

$$A + B = \begin{bmatrix} 2+1 & -1+5 \\ 3-2 & 4+0 \end{bmatrix} = \begin{bmatrix} 3 & 4 \\ 1 & 4 \end{bmatrix}$$

$$2A = \begin{bmatrix} 4 & -2 \\ 6 & 8 \end{bmatrix} \text{ and } 3B = \begin{bmatrix} 3 & 15 \\ -6 & 0 \end{bmatrix}$$

$$A + B = \begin{bmatrix} 3 & 4 \\ 1 & 4 \end{bmatrix} \text{ and } 2A - 3B = \begin{bmatrix} 1 & -17 \\ 12 & 8 \end{bmatrix}$$

**Example 2.2** Find the matrix  $X$  if  $X + \begin{bmatrix} 4 & 1 \\ -2 & 3 \end{bmatrix} = \begin{bmatrix} 6 & 0 \\ 1 & 5 \end{bmatrix}$ .

SOLUTION

$$X = \begin{bmatrix} 6 & 0 \\ 1 & 5 \end{bmatrix} - \begin{bmatrix} 4 & 1 \\ -2 & 3 \end{bmatrix}$$

$$X = \begin{bmatrix} 2 & -1 \\ 3 & 2 \end{bmatrix}$$

## 3 Multiplication of matrices

- The product  $AB$  exists only if the number of **columns of  $A$**  equals the number of **rows of  $B$** .
- If  $A$  is  $m \times n$  and  $B$  is  $n \times p$ , then  $AB$  is  $m \times p$ .
- The element in row  $i$  and column  $j$  of  $AB$  is obtained by multiplying the elements of row  $i$  of  $A$  by the corresponding elements of column  $j$  of  $B$  and adding.

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} p & q \\ r & s \end{bmatrix} = \begin{bmatrix} ap + br & aq + bs \\ cp + dr & cq + ds \end{bmatrix}$$

**Example 3.1** If  $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$  and  $B = \begin{bmatrix} 2 & 0 \\ 1 & 5 \end{bmatrix}$ , find  $AB$  and  $BA$ . Is  $AB = BA$ ?

SOLUTION

$$AB = \begin{bmatrix} 1 \times 2 + 2 \times 1 & 1 \times 0 + 2 \times 5 \\ 3 \times 2 + 4 \times 1 & 3 \times 0 + 4 \times 5 \end{bmatrix} = \begin{bmatrix} 4 & 10 \\ 10 & 20 \end{bmatrix}$$

$$BA = \begin{bmatrix} 2 \times 1 + 0 \times 3 & 2 \times 2 + 0 \times 4 \\ 1 \times 1 + 5 \times 3 & 1 \times 2 + 5 \times 4 \end{bmatrix} = \begin{bmatrix} 2 & 4 \\ 16 & 22 \end{bmatrix}$$

$AB \neq BA$ . Matrix multiplication is not commutative in general.

**Example 3.2** If  $A = \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}$ , show that  $A^2 - 5A + 7I = O$ .

SOLUTION

$$A^2 = \begin{bmatrix} 9-1 & 3+2 \\ -3-2 & -1+4 \end{bmatrix} = \begin{bmatrix} 8 & 5 \\ -5 & 3 \end{bmatrix}$$

$$5A = \begin{bmatrix} 15 & 5 \\ -5 & 10 \end{bmatrix} \text{ and } 7I = \begin{bmatrix} 7 & 0 \\ 0 & 7 \end{bmatrix}$$

$$A^2 - 5A + 7I = \begin{bmatrix} 8-15+7 & 5-5+0 \\ -5+5+0 & 3-10+7 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = O$$

#### 4 Properties of multiplication

- In general  $AB \neq BA$ .
- $A(BC) = (AB)C$  and  $A(B+C) = AB+AC$ , whenever the products exist.
- $AI = IA = A$  for a square matrix  $A$  and the identity matrix  $I$  of the same order.
- $AB = O$  does **not** mean that  $A = O$  or  $B = O$ .
- $AB = AC$  does **not** mean that  $B = C$ .
- $(A+B)^2 = A^2 + AB + BA + B^2$ , which is not  $A^2 + 2AB + B^2$  unless  $AB = BA$ .

#### 5 Solving matrix equations

- First find the **order** of the unknown matrix from the orders of the other matrices.
- Write the unknown matrix with letters for its elements, multiply out, and compare corresponding elements.

**Example 5.1** Find the matrix  $X$  such that  $\begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} X = \begin{bmatrix} 7 \\ 11 \end{bmatrix}$ .

SOLUTION

The first matrix is  $2 \times 2$  and the product is  $2 \times 1$ , so  $X$  is  $2 \times 1$ . Let  $X = \begin{bmatrix} x \\ y \end{bmatrix}$ .

$$\begin{bmatrix} 2x+y \\ x+3y \end{bmatrix} = \begin{bmatrix} 7 \\ 11 \end{bmatrix}, \text{ so } 2x+y=7 \text{ and } x+3y=11$$

Solving:  $x=2, y=3$

$$X = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

## 6 Common mistakes

- Adding matrices of different orders.
- Multiplying corresponding elements to find  $AB$ . Use row by column.
- Assuming  $AB = BA$ , or expanding  $(A + B)^2$  as in ordinary algebra.
- Writing the order as columns  $\times$  rows.
- Using the number 1 in place of the identity matrix  $I$ , or 0 in place of the zero matrix  $O$ .

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